

DEPLOY HIGHLY AVAILABLE WEB APPLICATION USING AWS APPLICATION LOAD BALANCER(ALB)+AUTO SCALING GROUP(ASG)

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Project: Application Load Balancer+ Auto Scaling Group(High Availability)

Objective

- Login AWS
- Purpose

Architecture Diagram

- Structure
- Explanation

AWS Services Used

- Amazon EC2
- Launch Template
- Auto Scaling Group (ASG)
- Application Load Balancer (ALB)
- Target Group
- Security Group
- Amazon VPC

Resources Created

- VPC
- Launch Template
- Auto Scaling Group
- Target Group
- Load Balancer
- EC2 Security Group
- ALB Security Group

Configuration Details

- Launch Template
- Auto scaling Group
- Target Group
- Application Load Balancer
- User Data Script
- Project Workflow

Conclusion

- Testing Performed
- Output
- Learning Outcome

1. Objective:

- Login to AWS Console
- To deploy a highly available web application on multiple EC2 instances using an Application Load Balancer (ALB) and Auto Scaling Group (ASG), ensuring automatic traffic distribution and automatic scaling based on demand.

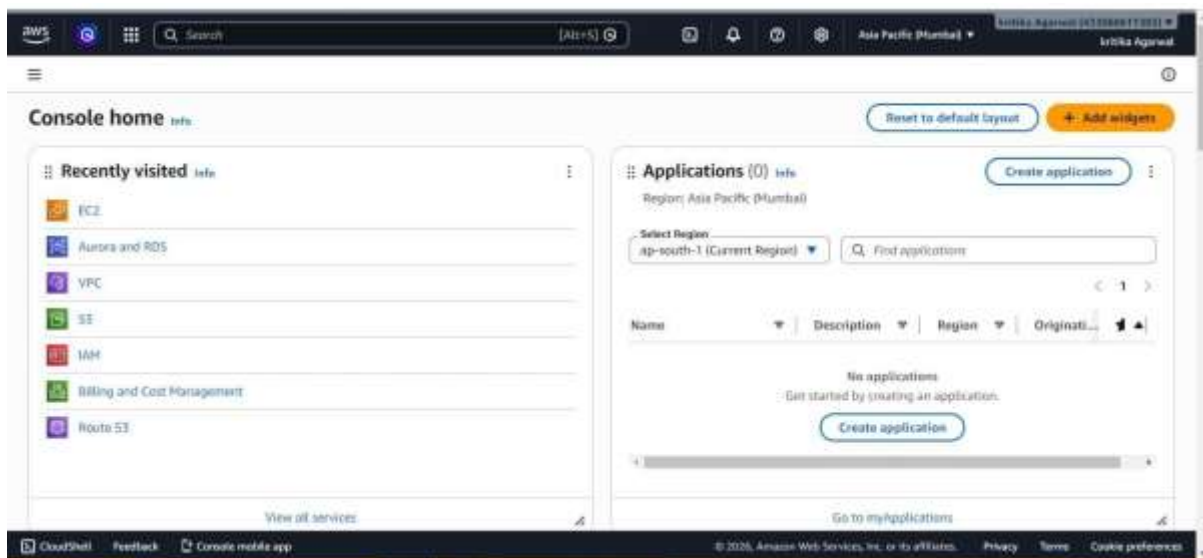


FIG: AWS CONSOLE

2. ARCHITECTURE DIAGRAM:

- Users: Users browser ke through website access karte hain.
- Application Load Balancer (ALB): Incoming traffic ko multiple EC2 instances me distribute karta hai.
- EC2 Instance 1 & EC2 Instance 2: Apache Web Server host karte hain aur user requests process karte hain.
- Auto Scaling Group (ASG): Traffic ke hisaab se automatically EC2 instances create ya terminate karta hai.
- Amazon RDS: Website ka relational database securely store karta hai aur sabhi EC2 instances database ko access kar sakte hain.

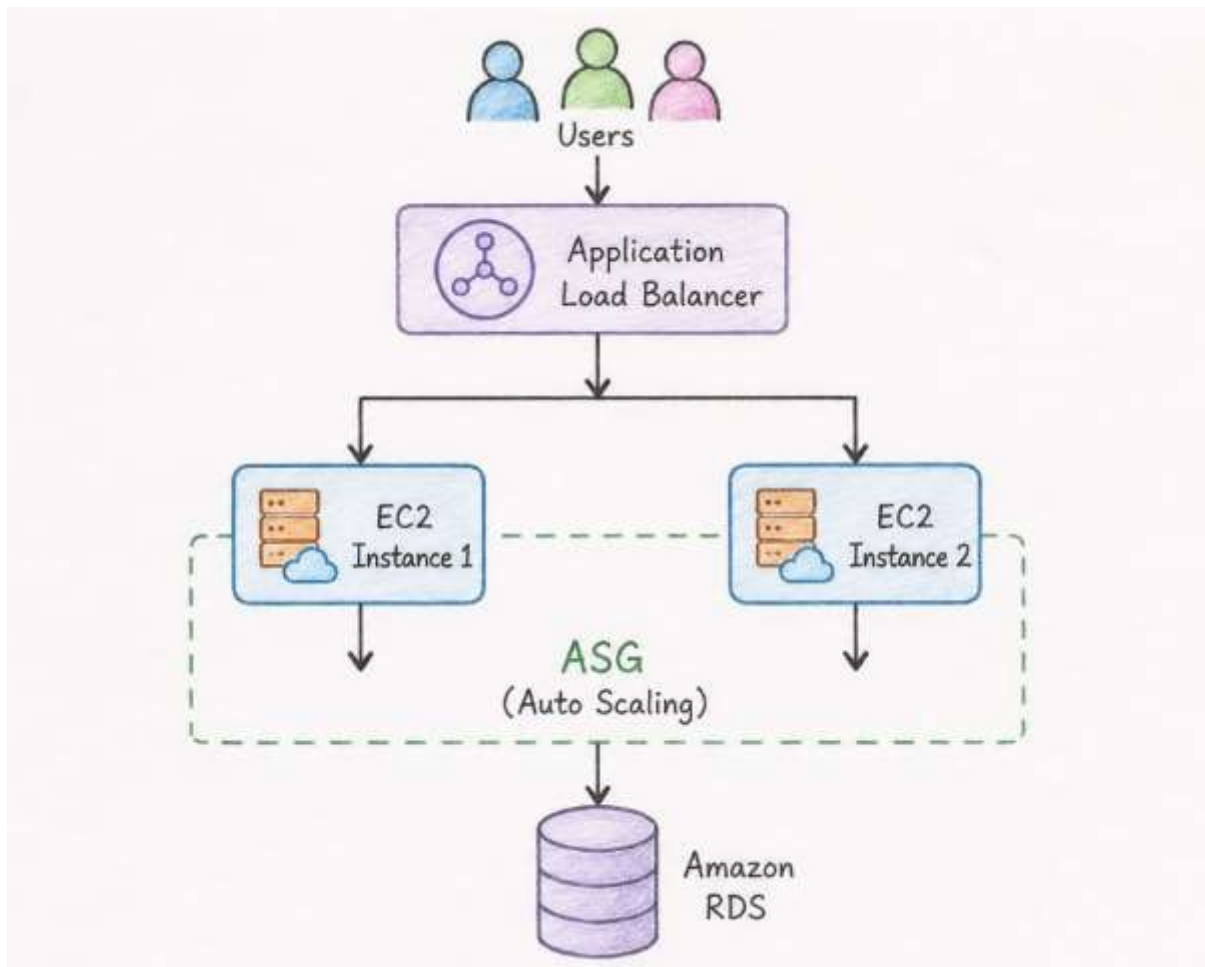


FIG: ARCHITECTURE DIAGRAM

3. AWS SERVICES USED:

1. Amazon EC2-

- Virtual server in the AWS Cloud.
- Application host karne ke liye use hota hai.
- CPU, RAM or Storage ke sath compute Service provide karta hai.

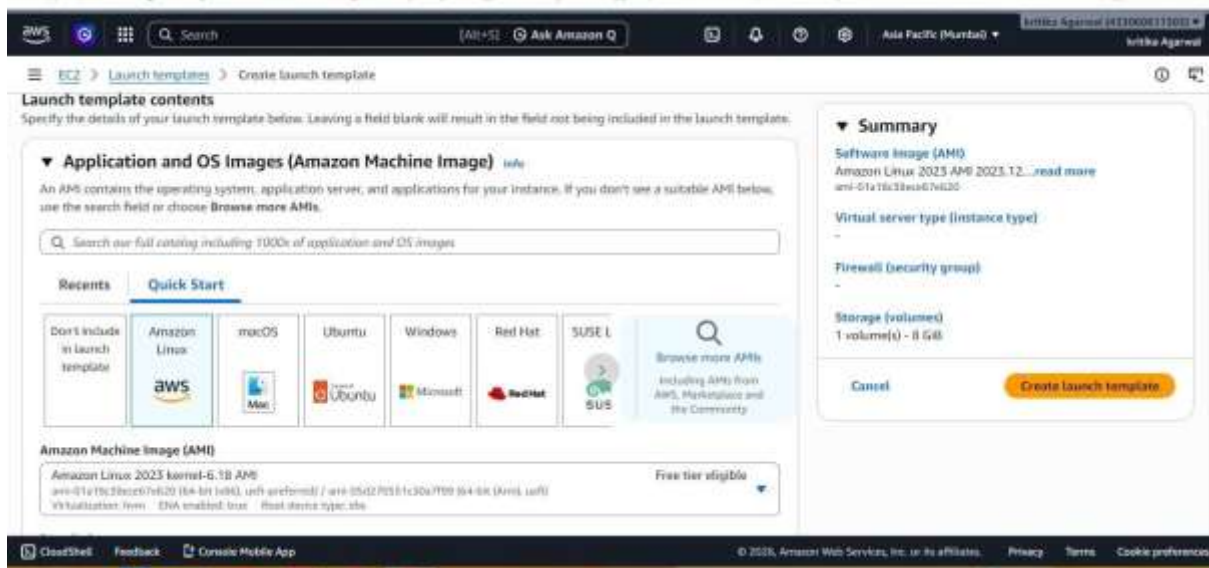


FIG: AMAZON EC2

2. Launch Template-

- EC2 launch karne ka blueprint hota hai.
- Isme AMI, Instance Type, Key Pair aur Security Group define hote hain.
- User Data script bhi add ki ja sakti hai.
- ASG isi template ka use karke naye instances create karta hai.

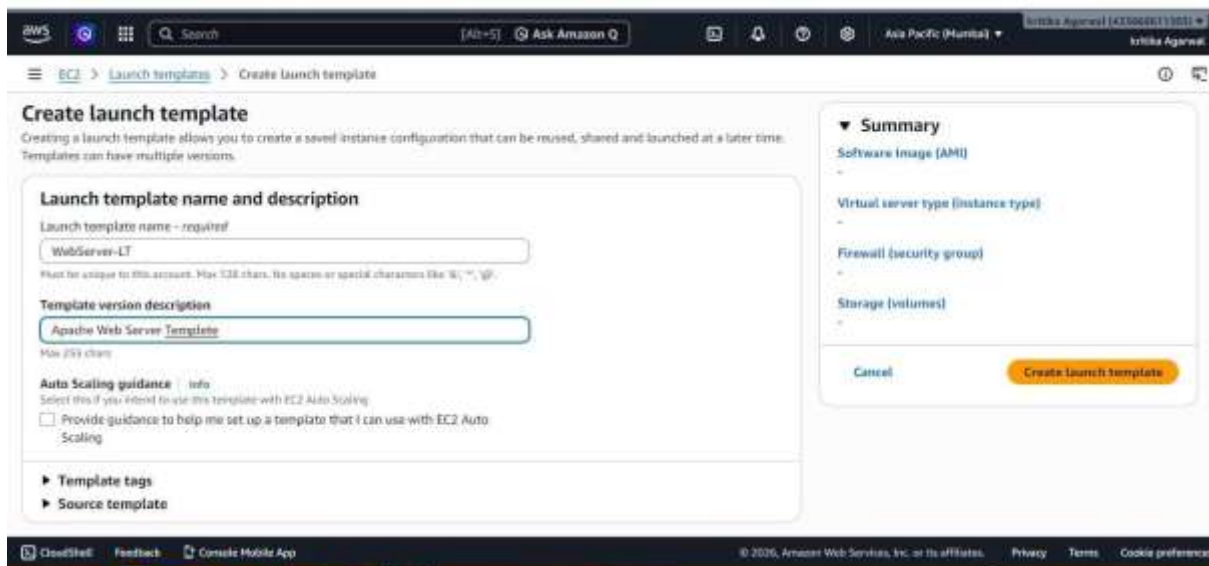


FIG-1: TEMPLATE CREATE

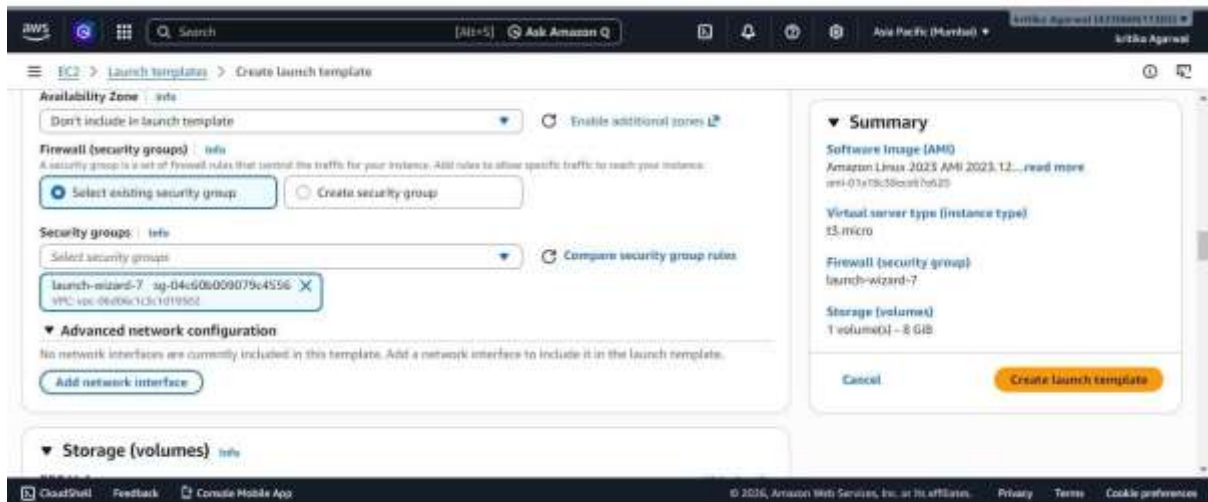


FIG-2: AZ/SG

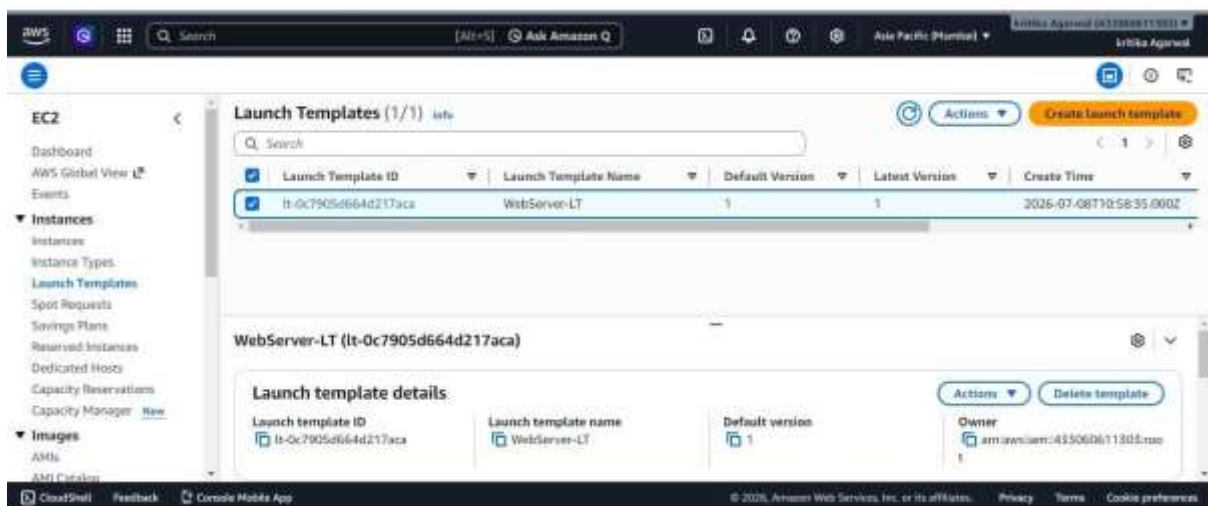


FIG-3: LAUNCH TEMPLATE

3. Auto Scaling Group (ASG)-

- Automatically EC2 instances manage karta hai.
- Traffic badhne par naye instances launch karta hai.
- Traffic kam hone par extra instances terminate karta hai.
- Desired, Minimum aur Maximum capacity maintain karta hai.



FIG-1:ASG(TEMPLATE NAME)

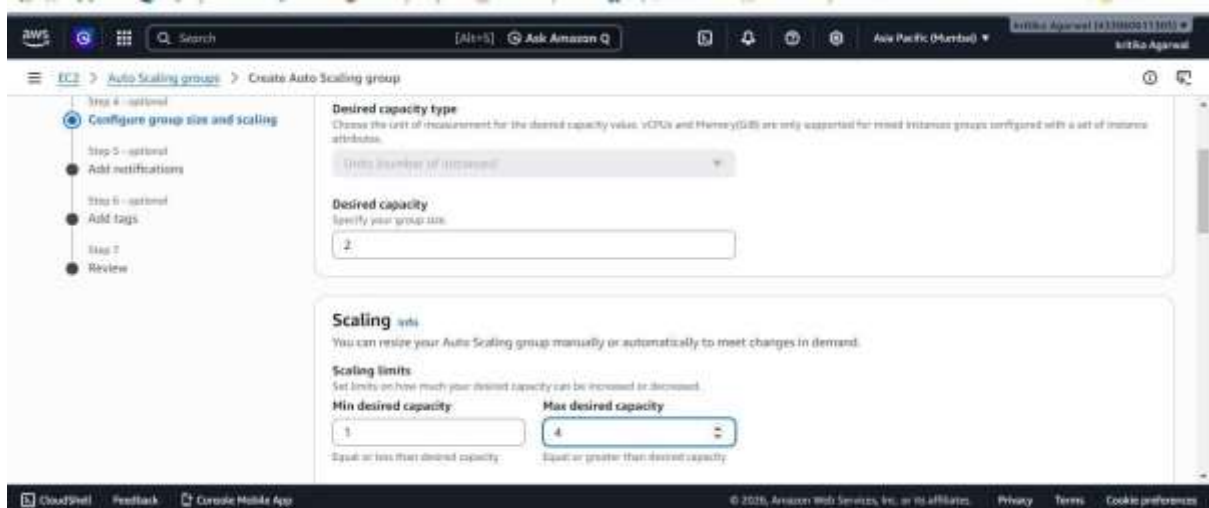


FIG-2: SCALING & SIZE

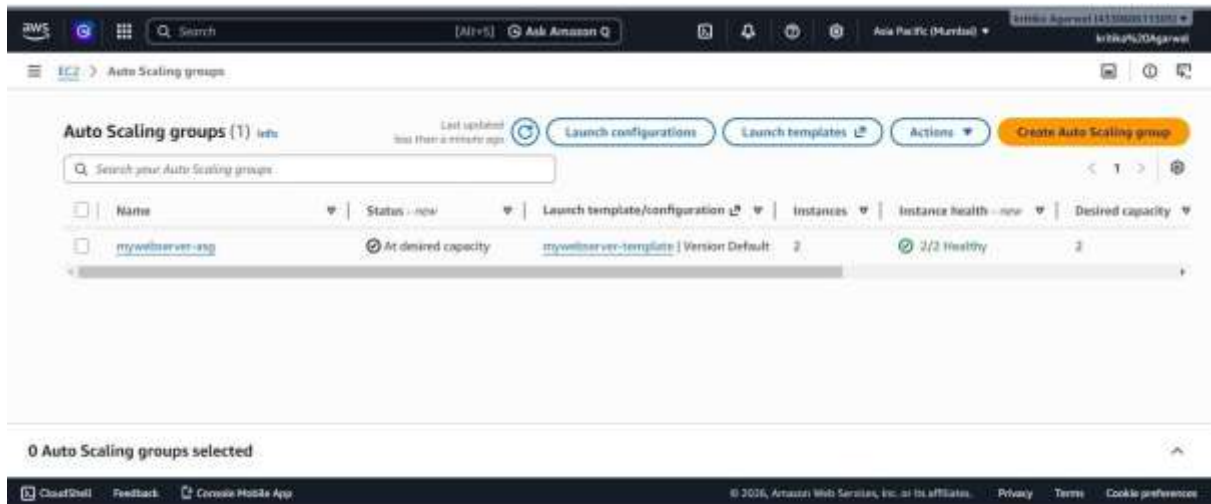


FIG-3: CREATE ASG

4. Application Load Balancer (ALB)-

- User requests ko multiple EC2 instances me distribute karta hai.
- Sirf Healthy instances ko traffic bhejta hai.
- High Availability provide karta hai.
- Website ko ek hi DNS Name se access karne deta hai.

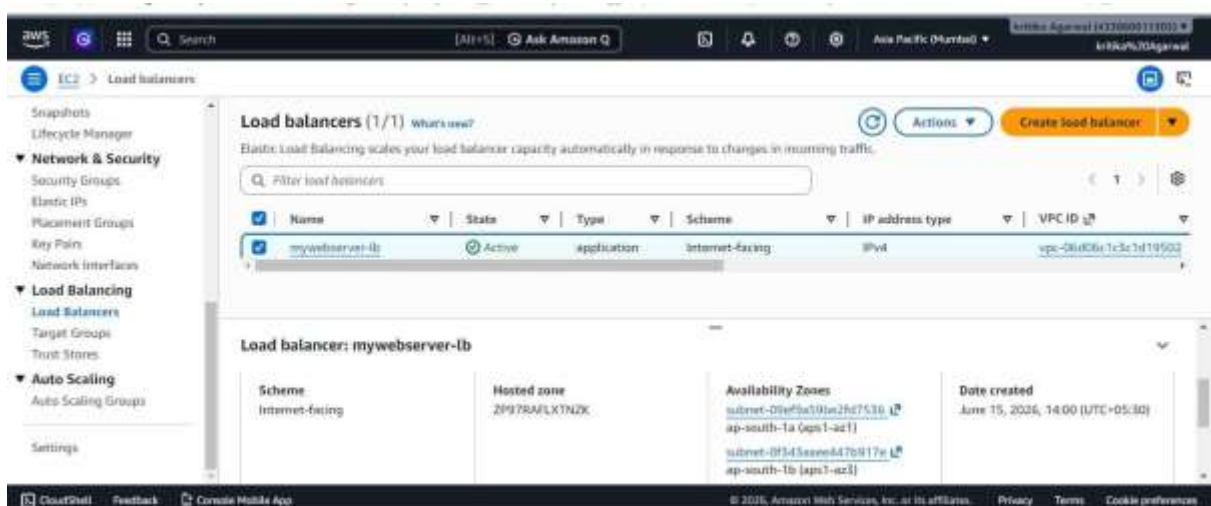


FIG-1: CREATE LB

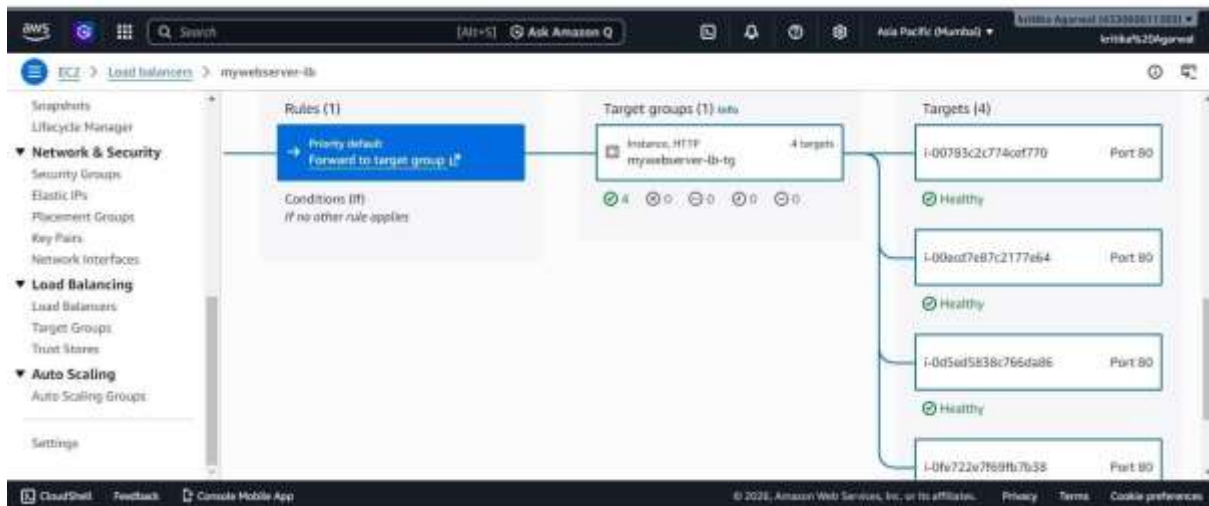


FIG-2: HEALTH CHECK

5.Target Group-

- EC2 instances ka group hota hai.
- ALB isi Target Group ko traffic forward karta hai.
- Health Checks perform karta hai.
- Sirf Healthy targets ko request milti hai.

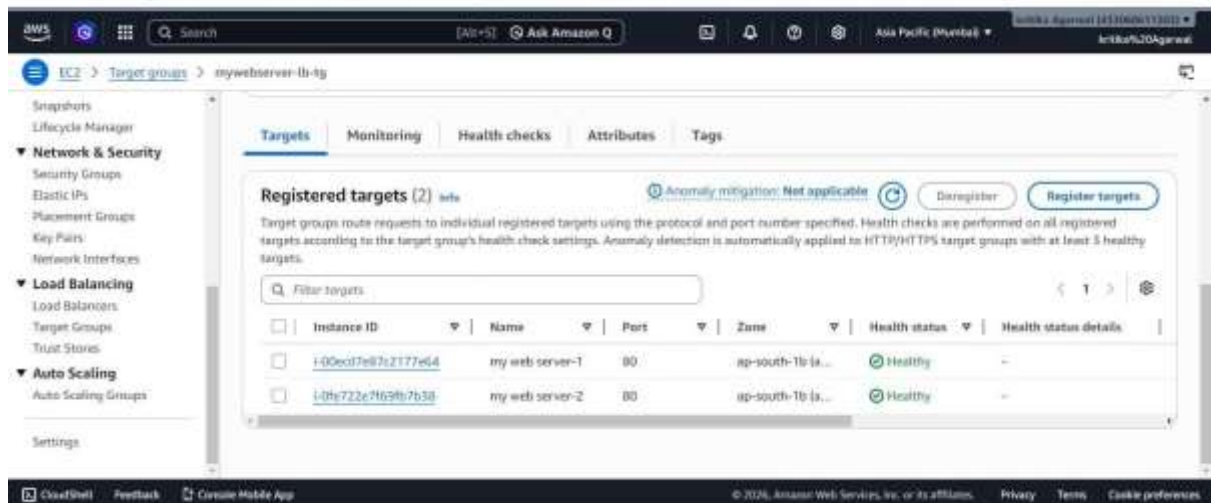


FIG-1: REGISTER TARGET



FIG-2: TARGET TYPE

6. Security Group-

- EC2 aur ALB ke liye Virtual Firewall hai.
- Inbound aur Outbound traffic control karta hai.
- HTTP (80), HTTPS (443) aur SSH (22) jaise ports allow karta hai.
- Resources ko unauthorized access se protect karta hai.

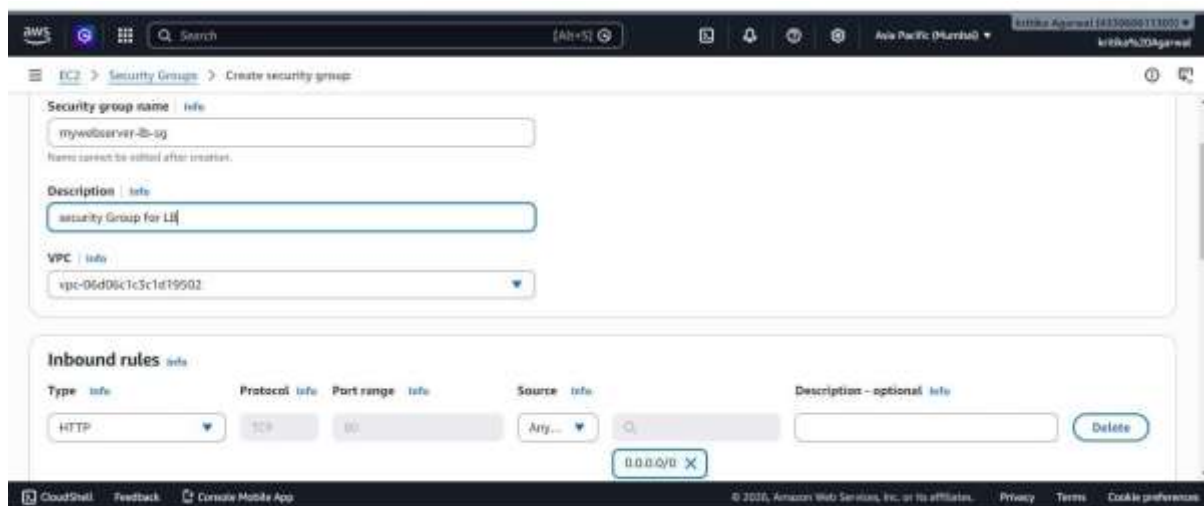


FIG-1: IN/OUT BOUND RULES

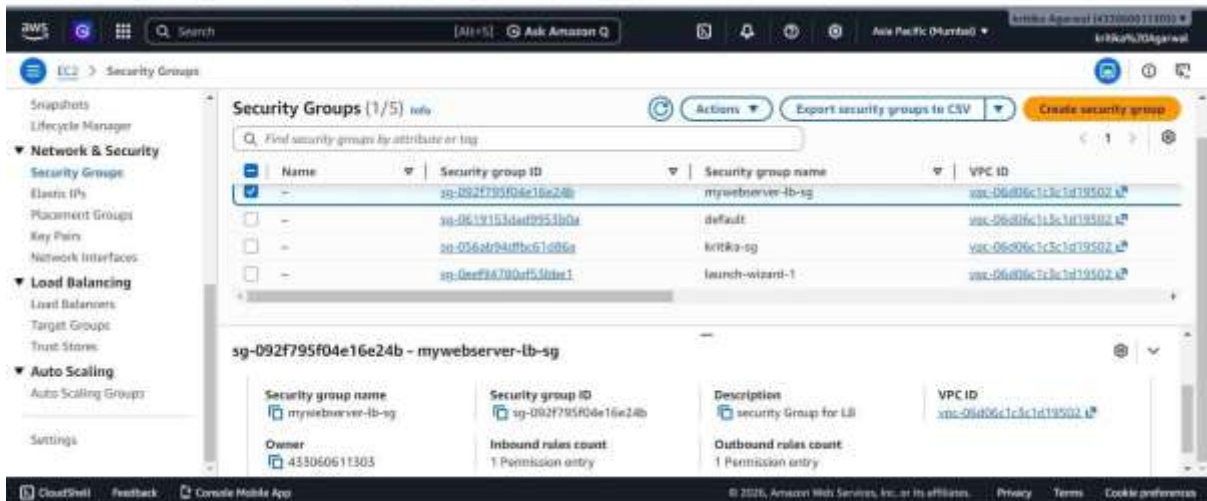


FIG-2: CREATE SG

7. Amazon VPC-

- AWS Cloud me private virtual network provide karta hai.
- Iske andar EC2, ALB aur RDS deploy hote hain.
- Subnets, Route Tables aur Internet Gateway manage karta hai.
- Network security aur isolation ensure karta hai.

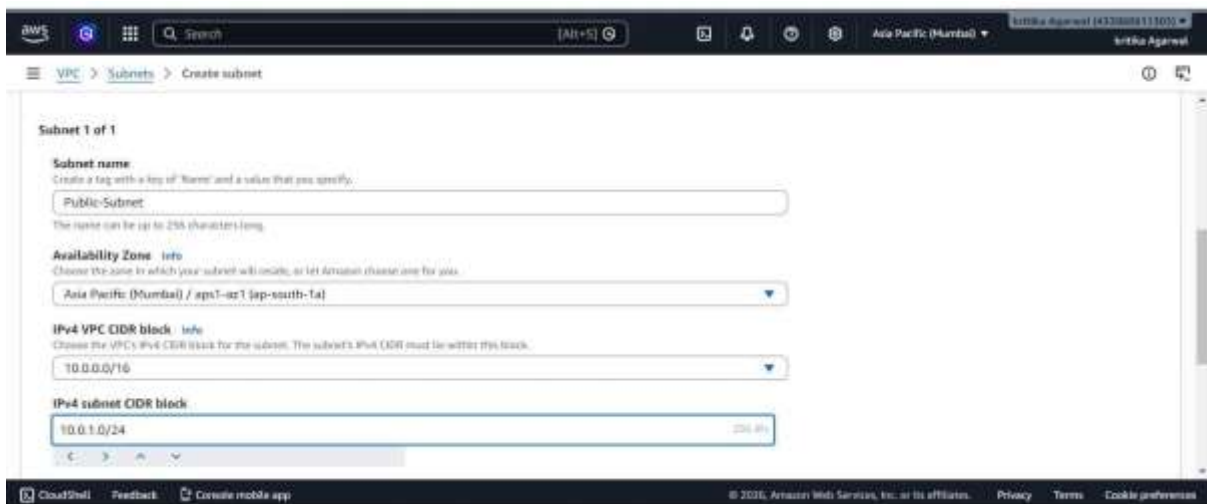


FIG-1: SUBNET

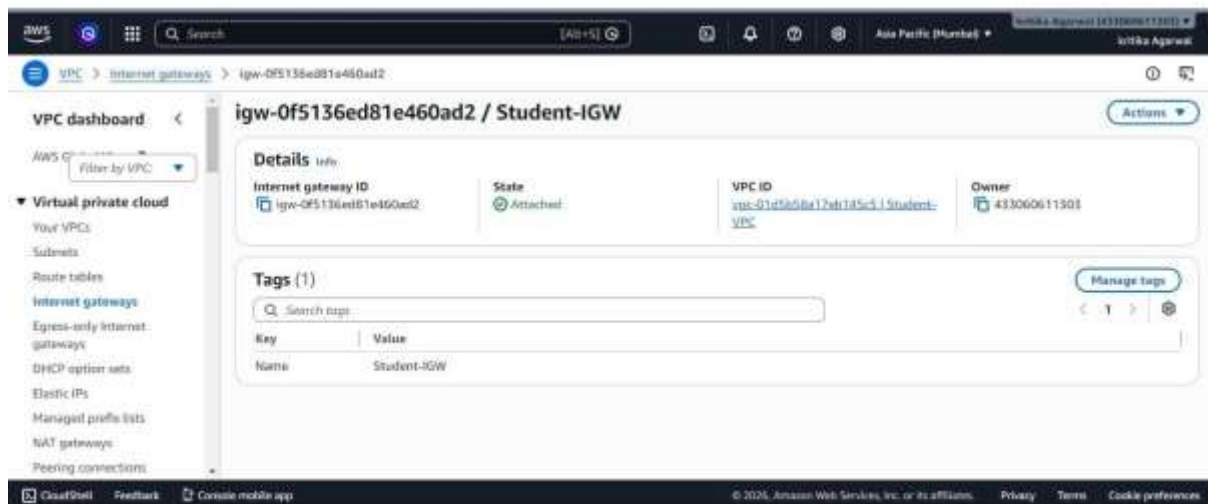


FIG-2: INTERNET GATEWAY

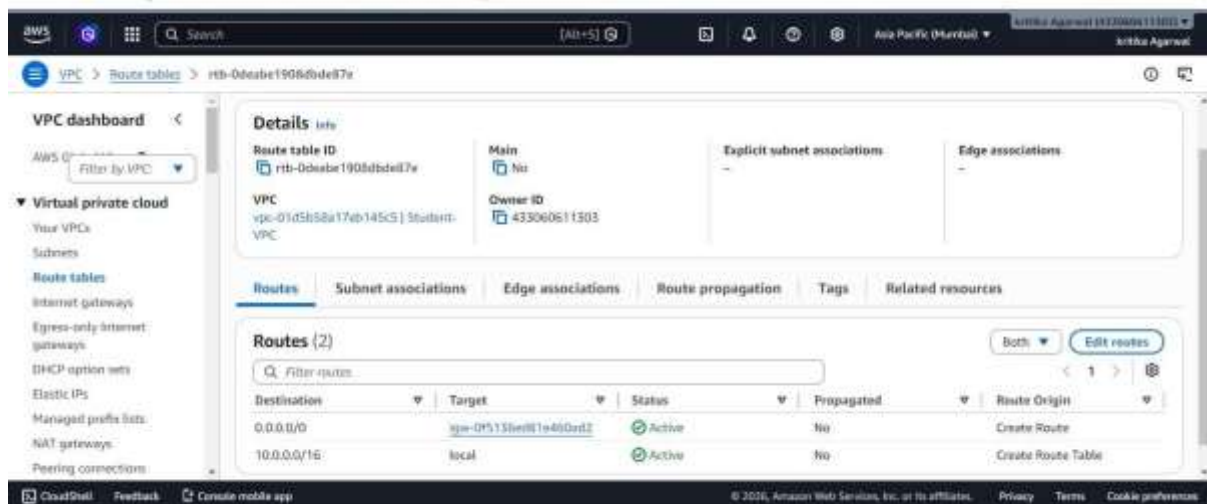


FIG-3:ROUTE TABLE

4. RESOURCES CREATED-

1. VPC-

- Created a private network for the project.

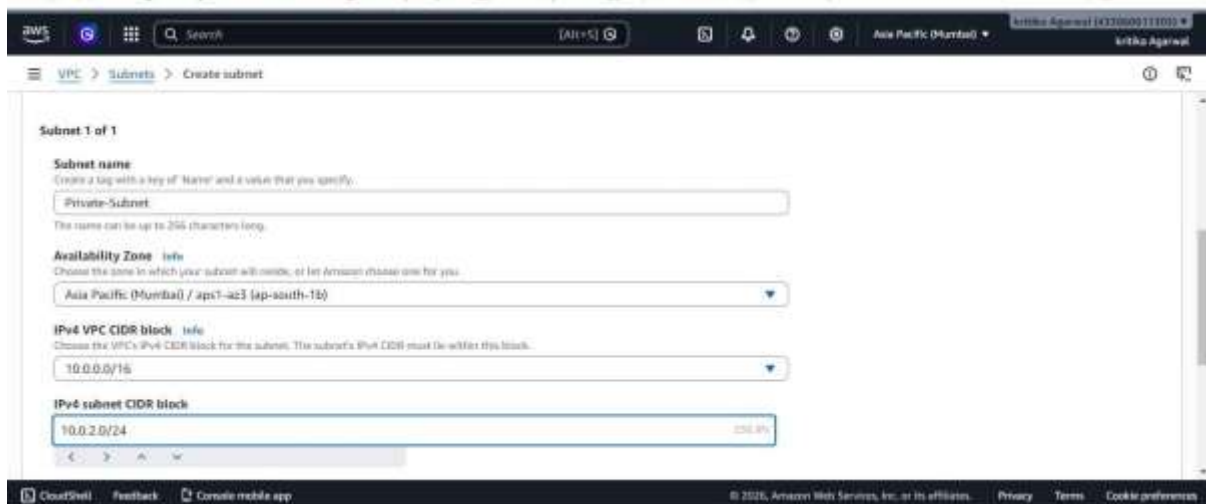


FIG: PRIVATE NETWORK

2. Launch Template –

- Created to define EC2 instance configuration.

3. Auto Scaling Group –

- Created to automatically launch and manage EC2 instances.

4. Target Group –

- Created to register EC2 instances for the ALB.

5. Application Load Balancer –

- Created to distribute incoming traffic.

6. EC2 Security Group –

- Created to allow SSH and HTTP traffic to EC2.

7. ALB Security Group –

- Created to allow HTTP traffic from the internet.

5. CONFIGURATION DETAILS-

1. Launch Template:

Explanation: Launch Template is used to define the configuration of EC2 instances that will be launched by the Auto Scaling Group.

Configuration:

- AMI: Amazon Linux 2023
- Instance Type: t3.micro
- Key Pair: My-Key

- Security Group: EC2-SG
- User Data: Apache installation script

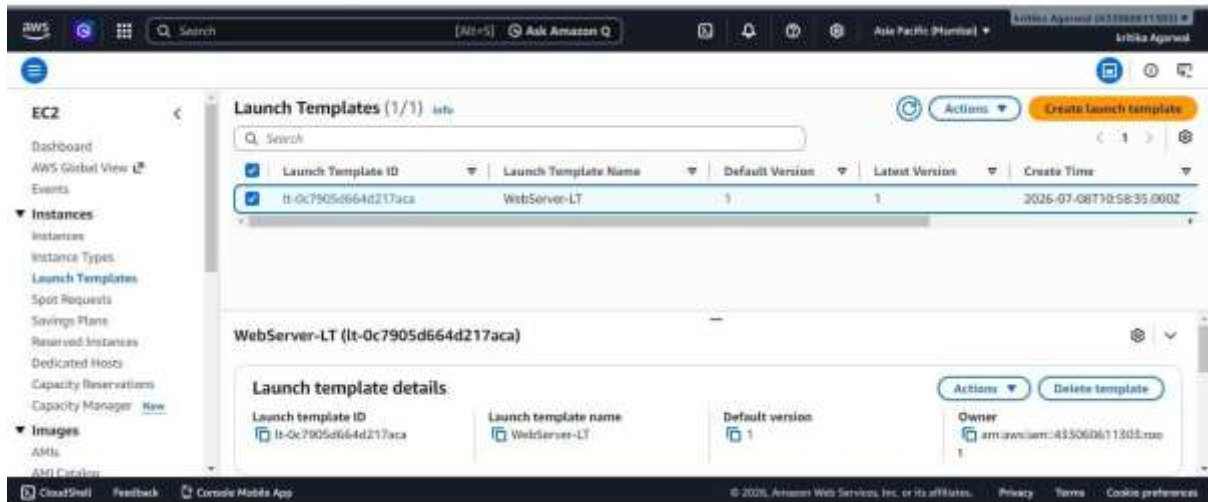


FIG: LAUNCH TEMPLATE

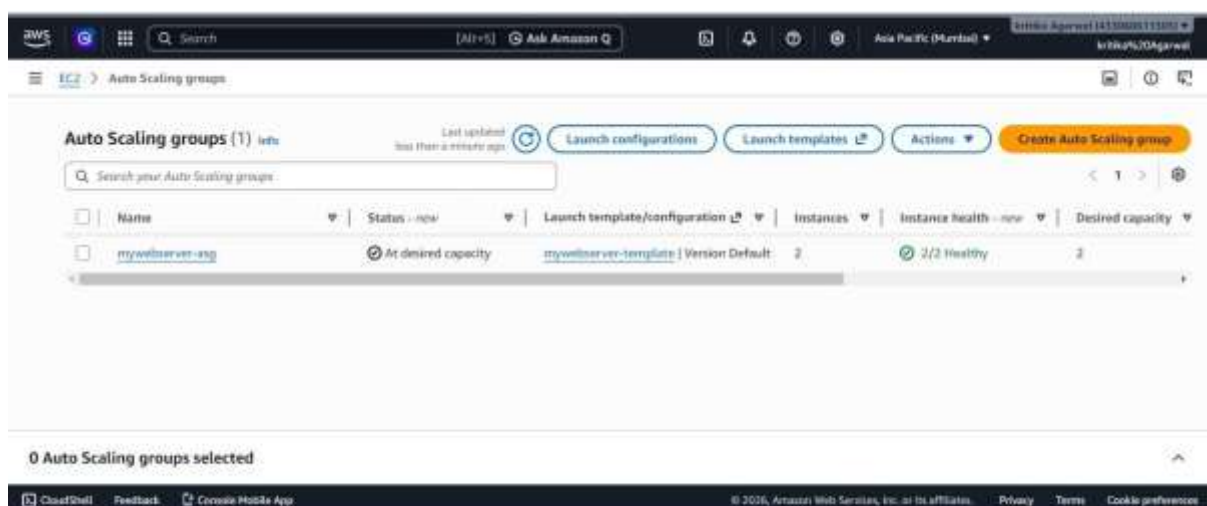
2. Auto Scaling Group (ASG)-

Explanation: Auto Scaling Group automatically launches or terminates EC2 instances based on the desired capacity and health status.

Configuration:

- Desired Capacity: 2
- Minimum Capacity: 1
- Maximum Capacity: 4
- Launch Template: WebServer-LT
- Target Group Attached: Yes
- Health Check Type: ELB
- Availability Zones: 2

FIG: AUTO SCALING GROUP



3. Target Group-

Explanation: Target Group registers EC2 instances and forwards requests received from the Application Load Balancer.

Configuration:

- Target Type: Instance
- Protocol: HTTP
- Port: 80
- Health Check Path: /

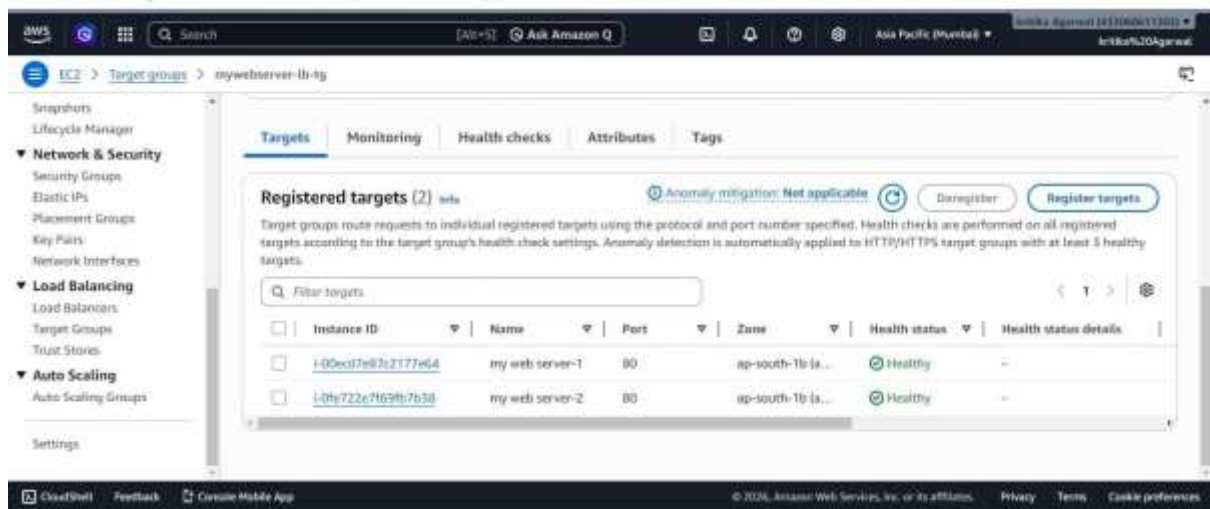


FIG:TARGET GROUP

4. Application Load Balancer (ALB)-

Explanation: Application Load Balancer distributes incoming HTTP requests among multiple EC2 instances to improve availability and reliability.

Configuration:

- Type: Internet-facing
- Listener: HTTP (Port 80)
- Target Group: Web-TG
- Availability Zones: 2
- Scheme: Public

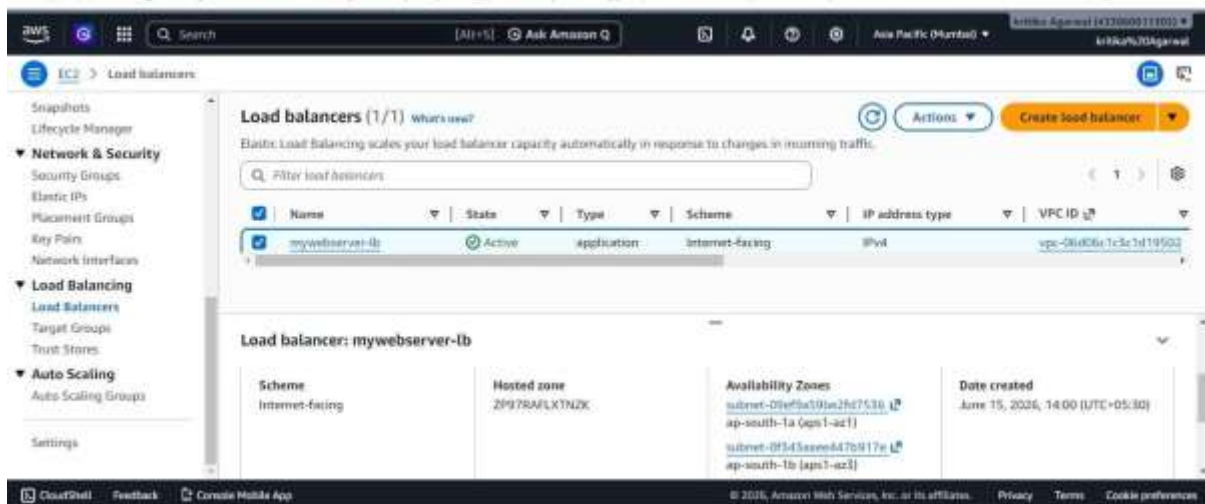


FIG: LOAD BALANCER

5. User Data Script-

Explanation: User Data Script automatically installs and starts Apache Web Server whenever a new EC2 instance is launched.

Example:-

```
#!/bin/bash
```

```
yum update -y
```

```
yum install httpd -y
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "<h1>Welcome to Auto Scaling Project</h1>" > /var/www/html/index.html
```

6. Project Workflow-

- User sends an HTTP request.
- Application Load Balancer receives the request.
- ALB forwards the request to the Target Group.
- Target Group selects a healthy EC2 instance.
- Auto Scaling Group maintains the required number of EC2 instances.
- If one instance fails, ASG automatically launches a new one.
- The user receives the response from the healthy EC2 instance.

6. CONCLUSION-

1. Testing Performed-

The following tests were successfully performed:

- Verified Launch Template configuration.
- Checked Auto Scaling Group instance creation.
- Verified Target Group health status.
- Tested Application Load Balancer DNS URL.
- Confirmed automatic traffic distribution between EC2 instances.
- Verified automatic replacement of unhealthy instances.

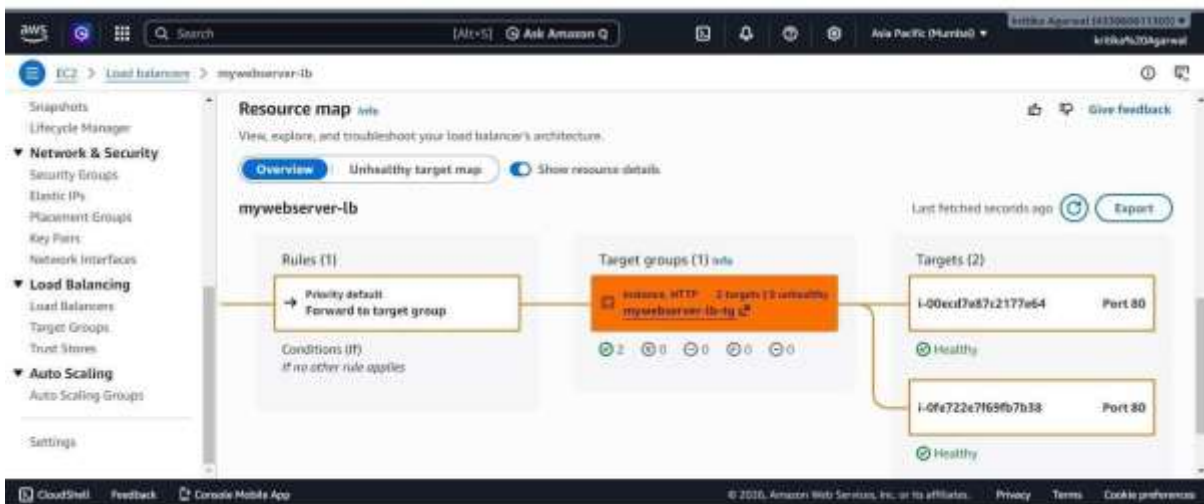


FIG-1: HEALTH STATUS

The screenshot shows the 'Instances' page in the AWS Management Console. It lists three EC2 instances. The instance 'my web server-2' is selected, and its details are shown below the table.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IP
My First Server	i-0971ab80f34cd0994	Stopped	t3.micro	—	ap-south-1a	—
my web server-1	i-00ecd7a87c2177e64	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-13-
my web server-2	i-0fe722e7f69fb7b38	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-13-

Details for 'my web server-2' (i-0fe722e7f69fb7b38):

- Instance ID: i-0fe722e7f69fb7b38
- Public IP address: 15.126.153.197
- Private IP addresses: 172.31.3.173

FIG-2: EC2 INSTANCE

2. Output-

- Website was successfully accessible through the ALB DNS name.
- Traffic was distributed across multiple EC2 instances.
- Auto Scaling automatically maintained the desired number of instances.
- High availability and fault tolerance were successfully achieved.



FIG-1: EC2 WEB SERVER



FIG-2: APACHE VERIFY

3. Learning Outcome-

After completing this project, I learned:

- How to create and configure a Launch Template.
- How to create and manage an Auto Scaling Group.
- How to configure an Application Load Balancer.

- How Target Groups and Health Checks work.
- How User Data automates EC2 setup.
- How to build a highly available and scalable web application architecture on AWS.

THANK YOU
